

Finding Thurstone: modeling Comparative Judgment data with R (and Stan)

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Abstract

The Classical Comparative Judgment (CCJ) analysis has become the standard approach for analyzing CJ data across multiple fields. Researchers favor it because it offers a simple approach for measuring traits and conducting analyses. Three features underpin its simplicity: it relies on restricted Thurstone-based models to estimate latent traits, it gives limited consideration to multiple assessment design features, and it separates trait measurement from analysis. Recent studies, however, question whether these modeling assumptions hold in contemporary CJ applications and whether the limited consideration of relevant design features, together with the separation of measurement and analysis, may bias estimates and reduce the precision of inferences.

To address these concerns, [Rivera et al. \(2025\)](#) proposed the Information-Theoretical Comparative Judgment (ITCJ) analysis, an umbrella analytical workflow designed to accommodate the complexity observed in applied CJ settings. The approach enables researchers to develop models tailored to the assumed CJ data-generating process, thereby reducing reliance on simplifying assumptions. In addition, by integrating measurement, analysis, and relevant assessment design features within a single framework, the approach facilitates coherent estimation and inference. However, these potential advantages require empirical validation.

Therefore, this study evaluates the ITCJ analysis using a simulated speech-quality dataset, benchmarking its performance against the CCJ analysis. The study also demonstrates the practical implementation of the approach, adopting a tutorial format with step-by-step guidance on data simulation, prior specification, model estimation, and interpretation using R, Stan, `cmdstan`, and `brms`. Ultimately, the study equips researchers with practical tools to apply the ITCJ analysis to more complex CJ assessments.

Keywords: tutorial, causal inference, bayesian inference, thurstone, comparative judgement

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1. Introduction

Comparative judgment (CJ) has emerged as a valuable methodology for measuring latent traits across diverse fields, including education (Kimbell, 2012; Jones and Inglis, 2015; van Daal et al., 2016; Bartholomew et al., 2018), psychology (Maydeu-Olivares and Böckenholt, 2005), political sciences (Zucco Jr. et al., 2019), linguistics (Boonen et al., 2020), and criminology (Seymour et al., 2022; Seymour and Hernandez, 2025). In CJ studies, judges actively compare pairs of stimuli to determine which stimulus exhibits more of the latent trait of interest (Thurstone, 1927b,a).

The *Classical Comparative Judgment* (CCJ) analysis has become the standard approach for analyzing CJ data across multiple fields, including education, linguistics, and psychology (see, for example, Wu, 2025; Thwaites and Paquot, 2024; Maydeu-Olivares and Böckenholt, 2005). Researchers favor this approach because it provides a convenient and simple method for measuring traits and conducting analyses (Andrich, 1978; Pollitt, 2012b). Three features underpin its simplicity. First, it relies on restricted Thurstone-based models to estimate latent traits. These models facilitate estimation by imposing simplifying assumptions about the traits, judges, and stimuli involved in the CJ assessment under study (Thurstone, 1927a; Bramley, 2008). Second, it pays limited attention to relevant CJ assessment design features that shape the data-generating process, such as the sampling and comparison mechanisms and potential judge biases (Rivera et al., 2025). Third, the approach separates trait estimation from follow-up analysis (Pollitt, 2012b), simplifying the latter by focusing only on mean trait estimates or residuals obtained from the former process (Rivera et al., 2025).

Recent studies, however, question whether the assumptions of these models hold in contemporary CJ applications, and whether the limited consideration of relevant assessment design features, together with the separation of trait estimation and analysis, may bias estimates and reduce the precision of inferences. (Bramley, 2008; Kelly et al., 2022; Rivera et al., 2025).

To address these concerns, Rivera et al. (2025) proposed the *Information-Theoretical Comparative Judgment* (ITCJ) analysis, an umbrella analytical workflow designed to accommodate the complexity often observed in applied CJ settings. The approach leverages causal and Bayesian inference methods to combine Thurstone’s core theoretical principles with relevant CJ assessment design features. This integration enables researchers to develop models tailored to the assumed data-generating process of the CJ system under study, thereby reducing reliance on simplifying assumptions. Moreover, by integrating measurement, analysis, and relevant design features within a single framework, the ITCJ analysis facilitates a single, coherent and flexible approach for trait estimation and statistical inference.

The flexibility of the ITCJ analysis is especially valuable in education, linguistics, and psychology,

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where CJ data often arise from complex assessment designs (Bramley, 2008; van Daal et al., 2017; Kelly et al., 2022) or involve comparisons of complex, heterogeneous stimuli (van Daal et al., 2016; Lesterhuis et al., 2018; Chambers and Cunningham, 2022). Its relevance, however, extends beyond these fields to any discipline using CJ assessments, as the approach allows researchers to model the processes they aim to study rather than those constrained by restrictive statistical models.

1.1. Research goals

The ITCJ analysis holds theoretical promise for producing reliable trait estimates and accurate statistical inferences (Rivera et al., 2025). However, as the authors acknowledge, this promise remains empirically untested. This study addresses this gap through two closely related research goals. First, it *empirically validates the ITCJ analysis* using a simulated speech-quality dataset, assessing the accuracy and reliability of its trait estimates and inferential parameters relative to the standard CCJ analysis.

Second, given the methodological complexity of the ITCJ approach, the study *demonstrates its practical implementation*, aiming to equip researchers with the tools needed to apply the ITCJ analysis to more complex CJ assessments. To this end, the study adopts a tutorial format, providing step-by-step guidance on data simulation, prior specification, model estimation, and interpretation using R (R Core Team, 2015), Stan (Stan Development Team., 2026b,a), and the interface packages `cmdstanr` (Gabry et al., 2025) and `brms` (Bürkner, 2017, 2018).

The remainder of this manuscript is organized into five sections. Section 2 reviews the standard analytical approach applied to CJ data: the CCJ analysis, and compares it with the ITCJ analysis. Section 3 describes the assumed data-generating process for the simulated dataset, the simulation procedure, the practical implementation of each analytical approach, and the evaluation criteria aligned with the research goals. Section 4 presents the data description and modeling results. Section 5 interprets the findings, outlines future research directions, and considers the study limitations. Finally, Section 6 offers the concluding remarks.

2. A tale of two analytical approaches

Pairwise comparison data are typically analyzed using the standard *CCJ analysis*. However, the *ITCJ analysis* proposed by Rivera and colleagues (2025) offers an alternative approach. This section provides an overview of both.

2.1. The CCJ analysis

The CCJ analysis typically proceeds through two sequential steps, each serving a distinct purpose: first, estimating traits, and second, conducting follow-up analyses (Pollitt, 2012a,b; Jones et al., 2019; Boonen et al., 2020; Chambers and Cunningham, 2022; Bouwer et al., 2023). Trait estimation

generally relies on Thurstone-based models, which yield two primary outputs: mean trait estimates and comparison-level residuals. For instance, [Boonen et al. \(2020\)](#) used the Bradley-Terry-Luce (BTL) model ([Bradley and Terry, 1952](#); [Luce, 1959](#)) to derive mean latent scores representing children’s overall speech quality, together with the corresponding comparison-level residuals.

Researchers then use these mean trait estimates and comparison-level residuals to conduct follow-up analyses. On the one hand, analyses of the mean latent scores fulfill several purposes, including summarizing estimates to aid interpretation, aggregating stimulus-level estimates to the individual level, partitioning variability between and within individuals, and drawing inferences about factors that influence trait scores. For example, [Maydeu-Olivares and Böckenholt \(2005\)](#) applied various factor models to latent means that represent individuals’ preferences for compact cars, to provide a more parsimonious representation of these preferences. Similarly, [Boonen et al. \(2020\)](#) applied a multilevel regression model to mean latent estimates to examine whether children’s age or hearing status affects their speech quality scores.

On the other hand, analyzing the residuals typically seeks to aggregate the remaining variability at the judge level, partition residual variability between and within judges, test for systematic biases, and identify potential misfitting stimuli, individuals, or judges. For instance, [Wu \(2025\)](#) fitted an analysis of variance (ANOVA) model to judges’ infit statistics to examine whether judge expertise influences judgment behavior. The infit statistics—a weighted average of the squared Pearson residuals—indicate the extent to which judges’ decisions deviate from expectations relative to those of other judges ([Wright and Masters, 1982](#)).

This two-step CCJ analysis has become standard practice in education, linguistics, and psychology (see, e.g., [Wu, 2025](#); [Thwaites and Paquot, 2024](#); [Maydeu-Olivares and Böckenholt, 2005](#)). Nevertheless, despite its widespread adoption, this approach has three key limitations: first, it relies on restricted Thurstone-based models to estimate traits; second, it often gives limited consideration to relevant CJ assessment design features; and third, it separates trait estimation from follow-up analysis.

Regarding the first limitation, although restricted Thurstone-based models facilitate trait estimation by imposing simplifying assumptions about traits, judges, and stimuli ([Thurstone, 1927a](#); [Bramley, 2008](#)), violations of these assumptions can undermine the reliability and validity of the resulting trait estimates ([Perron and Gillespie, 2015](#); [Rivera et al., 2025](#)). This concern is exemplified by [Wu \(2025\)](#), who, under the BTL model, implicitly treats second-language English texts as independent stimuli that are equally informative about individuals’ writing performance, despite their heterogeneous and complex nature ([Thurstone, 1927b](#); [Andrich, 1978](#); [Bramley, 2008](#); [Kelly et al., 2022](#)). Notably, if the texts differ in informativeness due to this heterogeneous nature, the BTL assumption of equal dispersions may be violated, potentially leading to an overestimation of the trait reliability

and misleading conclusions about differences between stimuli (McElreath, 2020; Wu et al., 2022). Moreover, if the characteristics of these texts unevenly influence judges’ perceptions, systematic biases may arise. Such biases can induce dependencies among stimuli, further contributing to an over- or underestimation of the trait reliability (van der Linden, 2017; Ackerman, 1989; Hoyle, 2023).

Concerning the second limitation, while giving limited attention to specific CJ assessment design features may also simplify trait estimation or follow-up analyses, Rivera and colleagues (2025) argued that neglecting elements that define the CJ data-generating process can obscure their broader role within the assessment system. The authors illustrated this concern by examining the sampling and comparison mechanisms. They showed that explicit inclusion of these mechanisms in the conceptual CJ model reveal their role as sources of missing data. They further argued that clarifying their function supports a more comprehensive evaluation of their impact on trait estimation, statistical inference, and the design of more complex assessment setups.

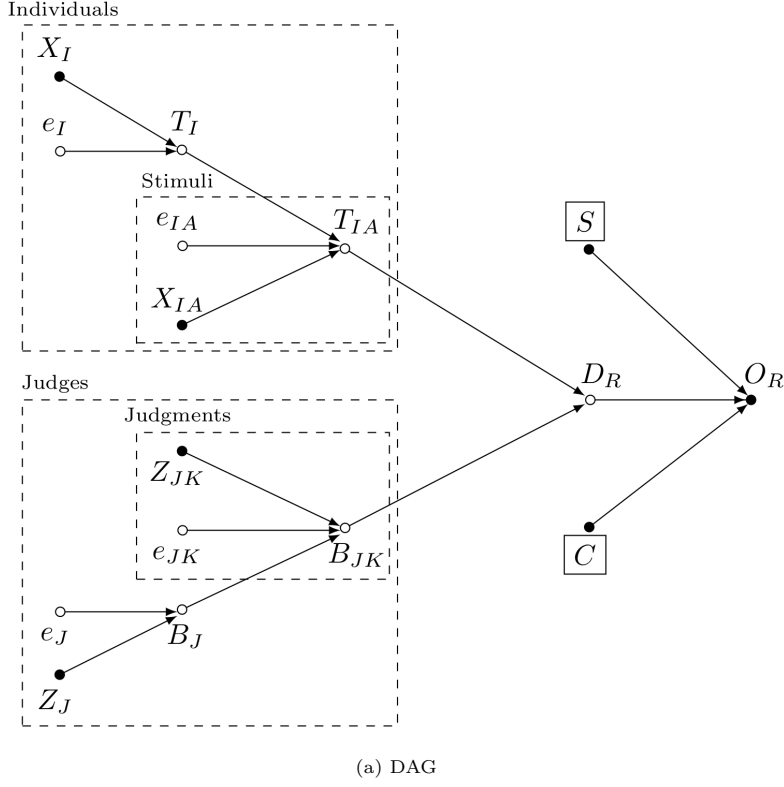
Lastly, concerning the third limitation, although separating trait estimation from follow-up analysis can simplify the latter, focusing solely on mean trait estimates or residuals obtained from the estimation stage inadvertently treats these outputs as fixed values rather than as uncertain parameter estimates (McElreath, 2020; Rivera et al., 2025). This separation prevents the propagation of uncertainty from the estimation stage to the follow-up analyses (Pritikin, 2020), which can bias parameter estimates and reduce their precision. The direction and magnitude of these biases are often difficult to predict. Effects may be attenuated, inflated, or remain unchanged depending on the level of uncertainty in the scores and the true effects under investigation (McElreath, 2020; Kline, 2023; Hoyle, 2023). In addition, reduced precision lowers statistical power and increases the risk of Type I and Type II errors (McElreath, 2020).

Several studies have made valuable attempts to incorporate specific features that address some of the limitations discussed above. However, most studies propose incremental models that address one or two issues at a time. As a result, other relevant components of the CJ process remain unrepresented, which may lead to model misspecification and produce biased or imprecise parameter estimates (Everitt and Skrandal, 2010). For example, Seymour et al. (2022) and Seymour and Hernandez (2025) extended the BTL model to incorporate spatially correlated stimuli and tied outcomes, applying these models to measure area-level deprivation and honour-based abuse, respectively. Although these extensions captured important aspects of the CJ data, the analysis did not explicitly model other potentially relevant factors, such as the sampling mechanism associated with their “convenience” data sample, possible judge biases and judge-specific effects, or hierarchical structures expected in the data, such as wards nested within counties. Similar patterns of underrepresented features appear in Pritikin (2020), Schramm and Batchelder (2021), and Gray et al. (2024), where models typically address one or two key design features or sources of heterogeneity at a time.

Notably, the *ITCJ analysis* proposed by Rivera and colleagues (2025) provides an alternative approach that addresses the limitations outlined above. In the following section, we summarize this approach, with particular emphasis on its differences from the CCJ analysis.

2.2. The ITCJ analysis

In contrast to the CCJ approach, the ITCJ analysis does not begin with the application of a predefined statistical model to the CJ data, from which primary outputs are extracted for subsequent analysis. Instead, it starts by explicitly representing the relationships among observed and latent variables that are thought to characterize the data-generating process of the CJ system under study. This structure is represented using a *Directed Acyclic Graph (DAG)* and an associated *Structural Causal Model (SCM)* (Morgan and Winship, 2014; Gross et al., 2018; Neal, 2020). On the basis of this causal representation, a *probabilistic model* tailored to the system under study is derived, from which one or more *statistical models* are subsequently formulated. These models are then used to estimate, summarize, and conduct statistical inferences on the traits of interest.



$$\begin{aligned}
O_R &:= f_O(D_R, S, C) \\
D_R &:= f_D(T_{IA}, B_{JK}) \\
T_{IA} &:= f_T(T_I, X_{IA}, e_{IA}) \\
T_I &:= f_T(X_I, e_I) \\
B_{JK} &:= f_B(B_J, Z_{JK}, e_{JK}) \\
B_J &:= f_B(Z_J, e_J) \\
\\
e_I &\perp \{e_J, e_{IA}, e_{JK}\} \\
e_J &\perp \{e_{IA}, e_{JK}\} \\
e_{IA} &\perp e_{JK}
\end{aligned}$$

(b) SCM

Figure 1: ITCJ analysis, plausible data-generating process for a CJ-based speech quality study

Figure 1 presents a DAG and SCM that represent a plausible data-generating process for a CJ-based speech quality assessment study. In DAG 1a, closed circles represent observed variables, whereas open circles represent latent (unobserved) variables. Arrows indicate the direction of the assumed influences between variables. Dashed boxes denote the assumed hierarchical structure of the data, such as stimuli nested within individuals. In contrast, SCM 1b represents the same structure using a

more formal mathematical notation, in which functions $f_{(\cdot)}$, known as *structural equations*, represent variables as non-parametric functions of other observed or latent variables. These equations use the symbol ‘ $:=$ ’ to denote the asymmetric direction of the assumed influences, and the symbol ‘ $\perp$ ’ to represent *d-separation*, a concept closely related to statistical (conditional) independence. [Rivera et al. \(2025\)](#) provides a more detailed explanation of these concepts.

Moreover, Figure 1 uses X_I to denote the set of variables believed to influence the individual traits T_I , i.e., the individuals’ ability to produce high-quality speech. Examples of such individual-level variables include hearing status and age—the latter serving as a proxy for language development ([Boonen et al., 2020](#)). Similarly, X_{IA} denotes the set of variables assumed to influence the stimulus traits T_{IA} , which represent the latent quality of the stimulus under comparison. Examples of such stimulus-level variables include the length of the speech sample and whether the sample originates from spontaneous speech or imitation. Notably, the stimulus traits T_{IA} correspond to what the CJ literature refers to as *discriminal processes*, which are thought to capture the psychological effect a stimulus exerts on the judge, or more simply, the judge’s perception of the stimulus quality ([Thurstone, 1927b,a](#)).

In a similar fashion, Z_J denotes the set of variables believed to influence the judges’ bias B_J , that is, the systematic deviation of a judge’s perception of a stimulus from the consensus. Examples of such judges-level variables include speech-judgment expertise and prior training in the CJ task. Similarly, Z_{JK} denotes the set of variables assumed to influence the judgment-level biases B_{JK} , which represent systematic deviations of an individual judgment from the judge-specific bias. Examples of such judgment-level variables include the time of day or day of the week at which the comparison was made, as well as the number of comparisons completed by a judge, used as a proxy for learning or fatigue effects during the CJ task. The error terms e_I , e_{IA} , e_J , and e_{JK} capture residual variation at each level.

Notably, because individual and stimulus traits are, in essence, not directly observable, [Thurstone \(1927b; 1927a\)](#) introduced the *Law of Comparative Judgment*. This law posits that, in pairwise comparisons, a judge perceives the stimulus with a *discriminal process* positioned further along the trait continuum as possessing more of the trait ([Bramley, 2008](#)). This suggests that pairwise comparison outcomes depend on relative, rather than absolute, positions on the continuum, implying that the observed CJ outcome is determined by the difference between the underlying discriminial processes (traits) of the stimuli ([Rivera et al., 2025](#)). Moreover, because discriminial processes become observable only through judges’ perceptions—and these perceptions may involve bias or systematic deviations from consensus (see [Pollitt and Elliott, 2003](#); [van Daal et al., 2016](#))—judge- and judgment-level biases are appropriately treated as integral components of the CJ system from the outset ([Rivera et al., 2025](#)). In this representation, these relationships are captured by the arrows from the T_{IA} and B_{JK} vectors to the *discriminal difference* D_R vector, and the corresponding

structural equation $D_R := f_D(T_{IA}, B_{JK})$.

Furthermore, Figure 1a represents the sampling and comparison mechanisms as two conditioned variables that determine how population outcomes become the “observed” sample outcomes (S and C , respectively shown with surrounding squares to denote conditioning). In this particular representation, these variables are depicted without incoming arrows, indicating that they are assumed to be independent of all other variables in the DAG. This, in turn, implies that S corresponds to a *simple random sampling (SRSg)* design, in which each repeated judgment, judge, stimulus, and individual has an equal probability of inclusion within their respective groups (Lawson, 2015), while C corresponds to a *random allocation comparative design* (Bramley, 2015), also known as an *incomplete block design* (Lawson, 2015), in which each repeated judgment has an equal probability of being included in the sample.

Finally, Figure 1a shows that the observed CJ outcome O_R is jointly influenced by the discriminial difference D_R and the sampling and comparison mechanisms (S and C , respectively). In typical CJ applications, O_R is often modeled as following a Bernoulli distribution (see Pollitt, 2012b; Bramley, 2008; Boonen et al., 2020; Seymour and Hernandez, 2025). However, the current DAG is intentionally agnostic to this specific distributional choice, thereby accommodating alternative outcome distributions as special cases within the ITCJ approach. This flexibility is particularly relevant in settings where the comparison task yields non-binary outcomes, for which extensions beyond the Bernoulli model may be more appropriate (Rivera et al., 2025). Examples of such outcomes can be found in Agresti (1992), and more recently in Pritikin (2020) and Lingel et al. (2024).

$$\begin{aligned}
& P(O_R, D_R, S, C, T_{IA}, X_{IA}, e_{IA}, T_I, X_I, e_I, B_{JK}, Z_{JK}, e_{JK}, B_J, Z_J, e_J) \\
&= P(O_R \mid D_R, S, C) \cdot P(S) \cdot P(C) \\
&\quad \cdot P(D_R \mid T_{IA}, B_{JK}) \\
&\quad \cdot P(T_{IA} \mid T_I, X_{IA}, e_{IA}) \cdot P(X_{IA}) \\
&\quad \cdot P(T_I \mid X_I, e_I) \cdot P(X_I) \\
&\quad \cdot P(B_{JK} \mid B_J, Z_{JK}, e_{JK}) \cdot P(Z_{JK}) \\
&\quad \cdot P(B_J \mid Z_J, e_J) \cdot P(Z_J) \\
&\quad \cdot P(e_I)P(e_{IA})P(e_J)P(e_{JK})
\end{aligned}$$

Figure 2: Probabilistic model, joint distribution (left hand side) factorized into conditional probability distributions (CPDs, right hand side).

Once researchers specify the SCM characterizing the assumed data-generating process of the CJ system, as in Figure 1b, they can translate it into a probabilistic model that factorizes the joint

distribution of the variables into a product of simpler *conditional probability distributions (CPDs)* (Pearl et al., 2016; Neal, 2020), as illustrated in Figure 2. From this formulation, they can then derive one or more statistical models tailored to the CJ system under study (see Section 3.4.2).

At this stage, it becomes clear that the ITCJ approach differs fundamentally from the CCJ approach in its treatment of trait estimation, summaries, and inference, ultimately enabling it to address the three main limitations of the latter. First, although the ITCJ analysis still relies on Thurstone-based models for trait estimation, these models are no longer treated as pre-specified simplifying abstractions; instead, they are adapted to the specific data-generating process of the CJ system under study, allowing researchers to explicitly model the processes of substantive interest. Second, the approach explicitly incorporates relevant features of CJ assessment design, including the sampling and comparison mechanisms as well as judge and judgment biases. Third, the ITCJ analysis no longer separates trait estimation from follow-up analyses, but rather integrates both within a unified and systematic approach (Rivera et al., 2025). This integration enables the propagation of uncertainty across all levels of the assumed hierarchical structure of the data.

Building on this framework, the following section details the methods used to implement and evaluate the ITCJ analysis against the CCJ approach.

3. Methods

To address its two research objectives, this study follows the *Bayesian research workflow* (Depaoli and van de Schoot, 2017; Pearl and Mackenzie, 2018; Neal, 2020; Gelman et al., 2020; Schad et al., 2020; Betancourt, 2020; McElreath, 2024a,b), as depicted in Figure 3. The workflow consists of iterative stages progressing from theoretical assumptions through parameter estimation and data prediction, thereby supporting the study’s dual focus on empirical model validation and tutorial-style implementation. Crucially, given the emphasis on validation, the study employs a synthetic dataset derived from a CJ data-generating process grounded in the speech-quality literature, enabling model validation under controlled conditions.

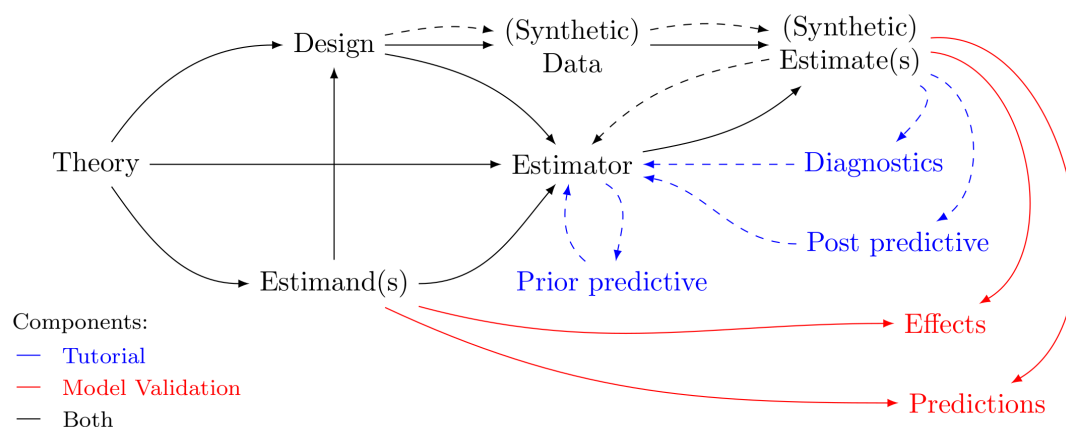


Figure 3: Bayesian research workflow.

The workflow begins by translating the theoretical understanding of the synthetic CJ system into formal *estimands*—the targets of inference (Everitt and Skrongdal, 2010), and uses these to guide the design of a data-generating process that reflects the stated assumptions. Specifically, drawing on the relevant literature, the study makes explicit the assumptions underlying the CJ structure of interest using a DAG and SCM, and then identifies and formally defines the estimands as target parameters.

The workflow then proceeds from theory through the design phase and into the data-generation phase. First, it simulates a synthetic conceptual population that reflects the specified theoretical assumptions. From this population, it draws a synthetic sample and corresponding comparison datasets (the latter constituting the *analysis dataset*).

Next, building on the theory and design stages, the ITCJ analysis translates the SCM into a probabilistic model. From this formulation, it derives one or more statistical models (*estimators*) tailored to the CJ system under study. For the CCJ analysis, the study adopts the BTL model (Bradley and Terry, 1952; Luce, 1959) as the estimator, given its extensive use in the CJ literature (recent examples include Wu, 2025; Thwaites and Paquot, 2024; Maydeu-Olivares and Böckenholt, 2005). The study then extracts trait estimates and residuals from the BTL model for use in subsequent follow-up analyses (see Section 3.4.1). At this phase, the study also conducts prior predictive checks to establish meaningful priors and to evaluate their implications within the model context before incorporating any empirical information (McElreath, 2020).

After specifying the estimators and their corresponding priors, the study fits the statistical models to obtain *estimates* of the target parameters under both approaches. Crucially, at this stage, the workflow evaluates these estimates using diagnostic and post-predictive assessments, including checks for stationarity, convergence, and mixing, as well as assessments of parameter recovery and predictive performance.

The following sections expand on specific methodological stages of the workflow.

3.1. From Theory to Design: Data-generating assumptions

The data-generating assumptions of this study draw on key features and findings from [Boonen et al. \(2020\)](#). In particular, the study also examines whether children’s age and hearing status influence overall speech quality, assuming a CJ task with the same core design features and empirical findings reported by the aforementioned authors.

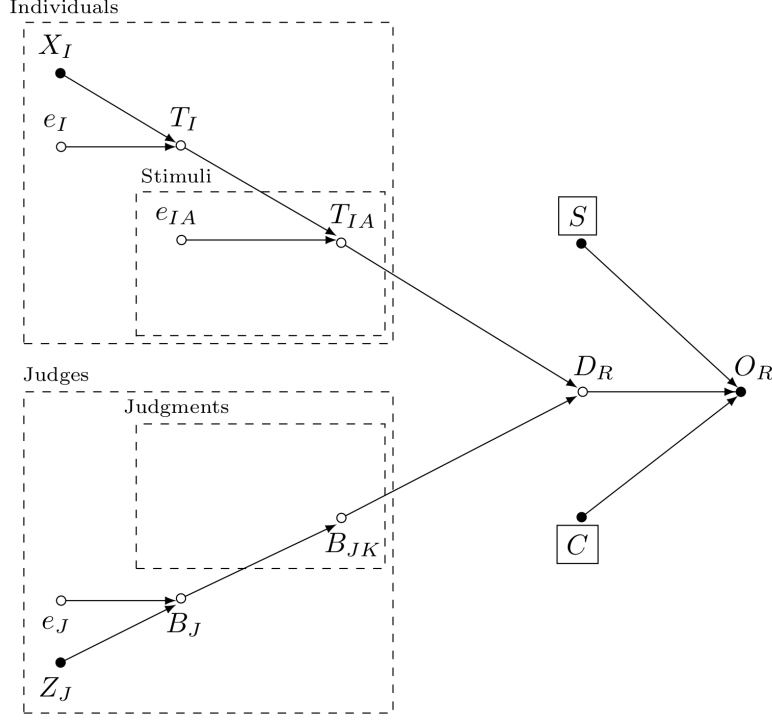


Figure 4: DAG of the “True” Data-Generating Process Underlying the Synthetic Conceptual Population

Figure 4 presents a DAG with the incorporated design features. The DAG represents the “true” data-generating process underlying the synthetic datasets. It depicts stimuli nested within individuals who differ in characteristics X_I , such as age and hearing status. It also includes judges who vary in speech-judgment expertise Z_J and who provide a single judgment per stimulus pair, reflected in the absence of variables Z_{JK} and e_{JK} previously shown in Figure 1a.

Finally, similar to Figure 1a, the DAG represents the sampling and comparison mechanisms (S and C , respectively) without incoming arrows, indicating their independence from all other variables in the DAG. This structure implies that the study samples both stimuli and individuals using SRSg algorithms ([Lawson, 2015](#)) and assigns judges to comparisons using random allocation comparative designs ([Bramley, 2015](#)) to generate the synthetic sampling and comparison datasets, respectively.

$O_R := f_O(D_R, S, C)$	$P(O_R \mid D_R, S, C)$	$O_R \stackrel{iid}{\sim} \text{Bernoulli}[\text{inv_logit}(D_R)]$
$D_R := f_D(T_{IA}, B_{JK})$	$P(D_R \mid T_{IA}, B_{JK})$	$D_R = (T_{IA}[i, a] - T_{IA}[h, b]) + B_{JK}[j, k]$
$T_{IA} := f_T(T_I, e_{IA})$	$P(T_{IA} \mid T_I, e_{IA})$	$T_{IA} = T_I + e_{IA}$
$T_I := f_T(X_I, e_I)$	$P(T_I \mid X_I, e_I)$	$T_I = \beta_{XI}X_I + e_I$
$B_J := f_B(Z_J, e_J)$	$P(B_J \mid Z_J, e_J)$	$B_J = \beta_{ZJ}Z_J + e_J$
$e_{IA} \perp \{e_I, e_J\}$	$P(e_{IA})P(e_I)P(e_J)$	$e \sim \text{Multi-Normal}(\mu, \Sigma)$
$e_I \perp \{e_J\}$		$\Sigma = VQV$
(a) SCM	(b) Probabilistic model	(c) Statistical model

Figure 5: SCM, Probabilistic and Statistical model of the “True” Data-Generating Process Underlying the Synthetic Conceptual Population

Next, the study incorporates Boonen and colleagues’ empirical findings through a statistical model derived from the probabilistic model implied by the system’s SCM, as depicted in Figure 5. The model conditions on the individual-level variable vector $X_I = \{X_{Ic}, X_{Id}[g = 1], X_{Id}[g = 2], X_{Id}[g = 3]\}$ with corresponding effect vector $\beta_{XI} = \{\beta_{XIc}, \beta_{XId[g=1]}, \beta_{XId[g=2]}, \beta_{XId[g=3]}\}$, as well as the judge-level variable vector $Z_J = \{Z_{Jd}[g = 1], Z_{Jd}[g = 2], Z_{Jd}[g = 3]\}$ with corresponding effect vector $\beta_{ZJ} = \{\beta_{ZJd[g=1]}, \beta_{ZJd[g=2]}, \beta_{ZJd[g=3]}\}$.

Specifically, the study assumes that each one-unit increase in age X_{Ic} leads to a 0.1-logit increase in speech quality for children aged five to nine, as captured by $\beta_{XIc} = 0.1$. In addition, the model assumes a strict ordering in hearing-status effects, such that normal-hearing (NH) children ($X_{Id}[g = 1]$) exhibit higher speech quality than hearing-impaired children with hearing aids (HI-HA, $X_{Id}[g = 2]$), who in turn outperform hearing-impaired children with cochlear implants (HI-CI, $X_{Id}[g = 3]$). This ordering is reflected in the parameters $\beta_{XId[g=1]} = 1$, $\beta_{XId[g=2]} = 0$, and $\beta_{XId[g=3]} = -1$.

Moreover, the study assumes that judges do not introduce systematic bias on average: audiologists (AU, $Z_{Jd}[g = 1]$), primary teachers (PT, $Z_{Jd}[g = 2]$), and inexperienced listeners (IL, $Z_{Jd}[g = 3]$) all have zero mean effects, such that $\beta_{ZJd[g=1]} = \beta_{ZJd[g=2]} = \beta_{ZJd[g=3]} = 0$.

Furthermore, consistent with Boonen et al. (2020), the study assumes that trait variability differs across hearing-status groups. In particular, NH children exhibit higher variability ($s_{XId[g=1]} = 1.5$) than HI-HA and HI-CI children, both of whom are assumed to show lower and equal variability ($s_{XId[g=2]} = s_{XId[g=3]} = 0.75$).

In addition, the study assumes greater unexplained variability between individuals than within individuals, i.e., the stimulus level. Formally, the model specifies $s_I = (s_{XIId[g=1]} + s_{XIId[g=2]} + s_{XIId[g=3]})/3 = 1$, which exceeds stimulus-level variability $s_A = 0.2$.

TO DO

3.2. From Theory to Estimands: Data-generating assumptions

3.3. From Design to Data: Data simulation

3.4. From Estimator to Estimate(s): The models

3.4.1. The CCJ analysis

3.4.2. The ITCJ analysis

3.4.2.1. Model 1.

3.4.2.2. Model 2.

3.4.2.3. Model 3.

3.4.2.4. Model 4.

3.4.2.5. Model 5.

3.4.2.6. Model 6.

3.5. From Estimate(s) to Diagnostics and Posterior predictives: The evaluation criteria

4. Results

4.1. Data description

4.2. Data modeling

4.2.1. The CCJ analysis

4.2.2. The ITCJ analysis

4.2.2.1. Model 1.

4.2.2.2. Model 2.

4.2.2.3. Model 3.

4.2.2.4. Model 4.

4.2.2.5. Model 5.

4.2.2.6. Model 6.

4.2.2.7. Model comparison.

5. Discussion

5.1. Future research directions

5.2. Study limitations

6. Conclusion

Declarations

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Licence: All the code that is original to this study and not attributed to any other authors is copyrighted by [Jose Manuel Rivera Espejo](#) and released under the new [BSD-3-Clause](#) license.

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7. Appendix

7.1. Appendix A: Stationarity, converge and mixing

7.2. Appendix B: Misfit observations

7.3. Appendix C: Sample size calculations

References

- Ackerman, T., 1989. Unidimensional irt calibration of compensatory and noncompensatory multidimensional items. *Applied Psychological Measurement* 13, 113–127. doi:[10.1177/014662168901300201](https://doi.org/10.1177/014662168901300201).
- Agresti, A., 1992. Analysis of ordinal paired comparison data. *Journal of the Royal Statistical Society* 41, 287–297. URL: <https://www.jstor.org/stable/2347562>, doi:[10.2307/2347562](https://doi.org/10.2307/2347562).
- Andrich, D., 1978. Relationships between the thurstone and rasch approaches to item scaling. *Applied Psychological Measurement* 2, 451–462. doi:[10.1177/014662167800200319](https://doi.org/10.1177/014662167800200319).
- Bartholomew, S., Nadelson, L., Goodridge, W., Reeve, E., 2018. Adaptive comparative judgment as a tool for assessing open-ended design problems and model eliciting activities. *Educational Assessment* 23, 85–101. doi:[10.1080/10627197.2018.1444986](https://doi.org/10.1080/10627197.2018.1444986).
- Betancourt, M., 2020. Toward a principled bayesian workflow. URL: https://betanalpha.github.io/assets/case_studies/principled_bayesian_workflow.html. last accessed in 10-08-2025.
- Boonen, N., Kloots, H., Gillis, S., 2020. Rating the overall speech quality of hearing-impaired children by means of comparative judgements. *Journal of Communication Disorders* 83, 1675–1687. doi:[10.1016/j.jcomdis.2019.105969](https://doi.org/10.1016/j.jcomdis.2019.105969).
- Bouwer, R., Lesterhuis, M., De Smedt, F., Van Keer, H., De Maeyer, S., 2023. Comparative approaches to the assessment of writing: Reliability and validity of benchmark rating and comparative judgement. *Journal of Writing Research* 15, 497–518. doi:[10.17239/jowr-2024.15.03.03](https://doi.org/10.17239/jowr-2024.15.03.03).
- Bradley, R., Terry, M., 1952. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika* 39, 324–345. doi:[10.2307/2334029](https://doi.org/10.2307/2334029).
- Bramley, T., 2008. Paired comparison methods, in: Newton, P., Baird, J., Goldsteing, H., Patrick, H., Tymms, P. (Eds.), *Techniques for monitoring the comparability of examination standards*. GOV.UK., pp. 246–300. URL: <https://assets.publishing.service.gov.uk/media/5a80d75940f0b62305b8d734/2007-comparability-exam-standards-i-chapter7.pdf>.
- Bramley, T., 2015. Investigating the reliability of adaptive comparative judgment. URL: <http://www.cambridgeassessment.org.uk/Images/232694-investigating-the-reliability-of-adaptive-comparative-judgment.pdf>. cambridge Assessment Research Report.
- Bürkner, P.C., 2017. brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software* 80, 1–28. doi:[10.18637/jss.v080.i01](https://doi.org/10.18637/jss.v080.i01).
- Bürkner, P.C., 2018. Advanced Bayesian multilevel modeling with the R package brms. *The R Journal* 10, 395–411. doi:[10.32614/RJ-2018-017](https://doi.org/10.32614/RJ-2018-017).
- Chambers, L., Cunningham, E., 2022. Exploring the validity of comparative judgement: Do judges attend to construct-irrelevant features? *Frontiers in Education* doi:[10.3389/educ.2022.802392](https://doi.org/10.3389/educ.2022.802392).
- Depaoli, S., van de Schoot, R., 2017. Improving transparency and replication in bayesian statistics: The wambs-checklist. *Psychological Methods* 22, 240–261. doi:[10.1037/met0000065](https://doi.org/10.1037/met0000065).
- Everitt, B., Skrondal, A., 2010. *The Cambridge Dictionary of Statistics*. Cambridge University Press.
- Gabry, J., Češnovar, R., Johnson, A., Brönders, S., 2025. cmdstanr: R Interface to 'CmdStan'. URL: <https://mc-stan.org/cmdstanr/>.
- Gelman, A., Vehtari, A., Simpson, D., Margossian, C.C., Carpenter, B., Yao, Y., Kennedy, L., Gabry, J., Bürkner, P.C., Modrák, M., 2020. Bayesian workflow. URL: <https://arxiv.org/abs/2011.01808>, [arXiv:2011.01808](https://arxiv.org/abs/2011.01808).
- Gray, A., Rahat, A., Crick, T., Lindsay, S., 2024. A bayesian active learning approach to comparative judgement within education assessment. *Computers and Education: Artificial Intelligence* 6, 100–245. URL: <https://www.sciencedirect.com/science/article/pii/S2666920X24000481>, doi:[10.1016/j.caeai.2024.100245](https://doi.org/10.1016/j.caeai.2024.100245).
- Gross, J., Yellen, J., Anderson, M., 2018. *Graph Theory and Its Applications*. Textbooks in Mathematics, Chapman and Hall/CRC. doi:<https://doi.org/10.1201/9780429425134>. 3rd edition.
- Hoyle, R.e., 2023. *Handbook of Structural Equation Modeling*. Guilford Press.
- Jones, I., Bisson, M., Gilmore, C., Inglis, M., 2019. Measuring conceptual understanding in randomised controlled trials:

- Can comparative judgement help? *British Educational Research Journal* 45, 662–680. doi:[10.1002/berj.3519](https://doi.org/10.1002/berj.3519).
- Jones, I., Inglis, M., 2015. The problem of assessing problem solving: can comparative judgement help? *Educational Studies in Mathematics* 89, 337–355. doi:[10.1007/s10649-015-9607-1](https://doi.org/10.1007/s10649-015-9607-1).
- Kelly, K., Richardson, M., Isaacs, T., 2022. Critiquing the rationales for using comparative judgement: a call for clarity. *Assessment in Education: Principles, Policy & Practice* 29, 674–688. doi:[10.1080/0969594X.2022.2147901](https://doi.org/10.1080/0969594X.2022.2147901).
- Kimbell, R., 2012. Evolving project e-scape for national assessment. *International Journal of Technology and Design Education* 22, 135–155. doi:[10.1007/s10798-011-9190-4](https://doi.org/10.1007/s10798-011-9190-4).
- Kline, R., 2023. *Principles and Practice of Structural Equation Modeling*. Methodology in the Social Sciences, Guilford Press.
- Lawson, J., 2015. *Design and Analysis of Experiments with R*. Chapman and Hall/CRC.
- Lesterhuis, M., van Daal, T., Van Gasse, R., Coertjens, L., Donche, V., De Maeyer, S., 2018. When teachers compare argumentative texts: Decisions informed by multiple complex aspects of text quality. *L1-Educational Studies in Language and Literature* 18, 1–22. doi:[10.17239/L1ESLL-2018.18.01.02](https://doi.org/10.17239/L1ESLL-2018.18.01.02).
- Lingel, H., Bürkner, P., Melchers, K., Schulte, N., 2024. Measuring personality when stakes are high: Are graded paired comparisons a more reliable alternative to traditional forced-choice methods? *Organizational Research Methods* 0, 10944281241279790. doi:[10.1177/10944281241279790](https://doi.org/10.1177/10944281241279790).
- Luce, R., 1959. On the possible psychophysical laws. *The Psychological Review* 66, 482–499. doi:[10.1037/h0043178](https://doi.org/10.1037/h0043178).
- Maydeu-Olivares, A., Böckenholt, U., 2005. Structural equation modeling of paired-comparison and ranking data. *Psychological Methods* 3, 285–304. doi:[10.1037/1082-989X.10.3.285](https://doi.org/10.1037/1082-989X.10.3.285).
- McElreath, R., 2020. *Statistical Rethinking: A Bayesian Course with Examples in R and STAN*. Chapman and Hall/CRC. doi:<https://doi.org/10.1201/9780429029608>.
- McElreath, R., 2024a. The third edition, the dotted half note of editions. URL: https://eleanth.org/2024_01_02_third_edition.html. last accessed in 10-08-2025.
- McElreath, R., 2024b. Understanding bayesian statistics without frequentist language. URL: <http://talmo.uk/2024/slides/mcelreath.pdf>. last accessed in 10-08-2025.
- Morgan, S., Winship, C., 2014. *Counterfactuals and Causal Inference: Methods and Principles for Social Research*. Analytical Methods for Social Research. 2 ed., Cambridge University Press.
- Neal, B., 2020. Introduction to causal inference from a machine learning perspective. URL: https://www.bradyn Neal.com/Introduction_to_Causal_Inference-Dec17_2020-Neal.pdf. last accessed 30 April 2024.
- Pearl, J., Glymour, M., Jewell, N., 2016. *Causal Inference in Statistics: A Primer*. John Wiley & Sons, Inc.
- Pearl, J., Mackenzie, D., 2018. *The Book of Why: The New Science of Cause and Effect*. 1st ed., Basic Books, Inc.
- Perron, B., Gillespie, D., 2015. Reliability and Measurement Error, in: *Key Concepts in Measurement*. Oxford University Press. Pocket guides to social work research methods. chapter 4. doi:[10.1093/acprof:oso/9780199855483.003.0004](https://doi.org/10.1093/acprof:oso/9780199855483.003.0004).
- Pollitt, A., 2012a. Comparative judgement for assessment. *International Journal of Technology and Design Education* 22, 157–170. doi:[10.1007/s10798-011-9189-x](https://doi.org/10.1007/s10798-011-9189-x).
- Pollitt, A., 2012b. The method of adaptive comparative judgement. *Assessment in Education: Principles, Policy and Practice* 19, 281–300. doi:[10.1080/0969594X.2012.665354](https://doi.org/10.1080/0969594X.2012.665354).
- Pollitt, A., Elliott, G., 2003. Finding a proper role for human judgement in the examination system. URL: <https://www.cambridgeassessment.org.uk/Images/109707-monitoring-and-investigating-comparability-a-proper-role-for-human-judgement.pdf>. research & Evaluation Division.
- Pritikin, J., 2020. An exploratory factor model for ordinal paired comparison indicators. *Heliyon* 6, e04821. doi:[10.1016/j.heliyon.2020.e04821](https://doi.org/10.1016/j.heliyon.2020.e04821).
- R Core Team, 2015. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna, Austria. URL: <http://www.R-project.org/>.
- Rivera, J., van Daal, T., De Maeyer, S., Gillis, S., 2025. Let’s talk about thurstone & Co.: an information-theoretical model for comparative judgments, and its statistical translation. URL: https://jriversaespejo.github.io/paper2_

- [manuscript/](#). last accessed in 30-08-2025.
- Schad, D.J., Betancourt, M., Vasisht, S., 2020. Toward a principled bayesian workflow in cognitive science. URL: <https://arxiv.org/abs/1904.12765>, [arXiv:1904.12765](#).
- Schramm, P., Batchelder, W.H., 2021. Hierarchical paired comparison modeling, a cultural consensus theory approach. *Journal of Mathematical Psychology* 100, 102465. URL: <https://www.sciencedirect.com/science/article/pii/S0022249620300973>, doi:<https://doi.org/10.1016/j.jmp.2020.102465>.
- Seymour, R.G., Hernandez, F., 2025. Scalable bayesian inference for bradley–terry models with ties: an application to honour based abuse. *Journal of Applied Statistics* 52, 1695–1712. doi:[10.1080/02664763.2024.2436608](https://doi.org/10.1080/02664763.2024.2436608).
- Seymour, R.G., Sirl, D., Preston, S.P., Dryden, I.L., Ellis, M.J.A., Perrat, B., Goulding, J., 2022. The bayesian spatial bradley–terry model: Urban deprivation modelling in tanzania. *Journal of the Royal Statistical Society Series C: Applied Statistics* 71, 288–308. URL: https://academic.oup.com/jrsssc/article-pdf/71/2/288/49158339/rssc_71_2_288.pdf, doi:[10.1111/rssc.12532](https://doi.org/10.1111/rssc.12532).
- Stan Development Team., 2026a. Stan Reference Manual, version 2.38.0. Vienna, Austria. URL: <https://mc-stan.org>.
- Stan Development Team., 2026b. Stan User's Guide, version 2.38.0. Vienna, Austria. URL: <https://mc-stan.org>.
- Thurstone, L., 1927a. A law of comparative judgment. *Psychological Review* 34, 482–499. doi:[10.1037/h0070288](https://doi.org/10.1037/h0070288).
- Thurstone, L., 1927b. Psychophysical analysis. *American Journal of Psychology* , 368–89URL: https://brocku.ca/MeadProject/Thurstone/Thurstone_1927g.html. last accessed 20 december 2024.
- Thwaites, P., Paquot, M., 2024. Comparative judgement for advancing research in applied linguistics. *Research Methods in Applied Linguistics* 3, 100142. URL: <https://www.sciencedirect.com/science/article/pii/S277276612400048X>, doi:<https://doi.org/10.1016/j.rmal.2024.100142>.
- van Daal, T., Lesterhuis, M., Coertjens, L., Donche, V., De Maeyer, S., 2016. Validity of comparative judgement to assess academic writing: examining implications of its holistic character and building on a shared consensus. *Assessment in Education: Principles, Policy & Practice* 26, 59–74. doi:[10.1080/0969594X.2016.1253542](https://doi.org/10.1080/0969594X.2016.1253542).
- van Daal, T., Lesterhuis, M., Coertjens, L., van de Kamp, M., Donche, V., De Maeyer, S., 2017. The complexity of assessing student work using comparative judgment: The moderating role of decision accuracy. *Frontiers in Education* 2. doi:[10.3389/feduc.2017.00044](https://doi.org/10.3389/feduc.2017.00044).
- van der Linden, W. (Ed.), 2017. Handbook of Item Response Theory: Statistical Tools. volume 2 of *Statistics in the Social and Behavioral Sciences Series*. CRC Press.
- Wright, B., Masters, G., 1982. Rating scale analysis. MESA Press. URL: <https://research.acer.edu.au/measurement/2>.
- Wu, Q., 2025. Comparative judgment: Building a shared consensus over rater variation in assessing second language writing performance. *Sage Open* 15, 21582440251346346. doi:[10.1177/21582440251346346](https://doi.org/10.1177/21582440251346346).
- Wu, W., Niezink, N., Junker, B., 2022. A diagnostic framework for the bradley–terry model. *Journal of the Royal Statistical Society Series A: Statistics in Society* 185, S461–S484. URL: https://academic.oup.com/jrsssa/article-pdf/185/Supplement_2/S461/49421054/jrsssa_185_supplement_2_s461.pdf, doi:[10.1111/rssa.12959](https://doi.org/10.1111/rssa.12959).
- Zucco Jr., C., Batista, M., Power, T.J., 2019. Measuring portfolio salience using the bradley–terry model: An illustration with data from brazil. *Research & Politics* 6, 2053168019832089. doi:[10.1177/2053168019832089](https://doi.org/10.1177/2053168019832089).